

MIDWAY ISLAND, United States Minor Outlying Islands

WMO: 910660

Lat: 28.202N

Lon: 177.383W

Elev: 6

StdP: 101.26

Time Zone: -11.00 (W11)

Period: 98-19

WBAN: 22701

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.3	15.3	5.4	5.6	15.9	6.8	6.1	16.7	14.7	19.1	13.1	19.0	4.1	340	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	31.1	25.5	30.2	25.1	29.6	24.7	26.3	29.2	25.8	28.7	25.5	28.5	4.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.5	20.7	27.9	24.9	20.0	27.7	24.5	19.4	27.5	80.9	29.2	78.9	28.9	77.1	28.5	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.9	10.7	9.2	DB	12.9	31.9	1.2	2.4	12.0	33.6	11.3	35.0	10.6	36.3	9.7	38.0	
			WB	9.0	27.2	0.8	1.0	8.5	27.9	8.0	28.5	7.6	29.0	7.0	29.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	19.7	19.3	19.9	21.0	22.4	25.1	26.6	27.5	27.1	25.3	23.1	21.3
	DBStd	3.35	2.05	2.18	2.03	1.82	2.05	1.54	1.00	1.13	1.25	1.62	1.70	1.80
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	34	10	13	8	2	0	0	0	0	0	0	0	1
	CDD10.0	4819	302	261	306	330	386	453	516	541	512	473	392	349
	CDD18.3	1813	53	42	56	81	128	203	257	283	262	214	142	93
	CDH23.3	11965	38	42	61	112	411	1377	2352	2956	2591	1483	426	118
	CDH26.7	2252	3	2	4	2	28	168	456	783	609	183	13	2
(14) Wind	WSAvg	4.8	5.8	5.4	5.1	4.8	4.4	4.3	4.8	4.5	4.0	4.5	5.1	5.1
(15) Precipitation	PrecAvg	1049	135	90	96	63	41	51	97	118	109	92	115	108
	PrecMax	1477	395	205	257	180	182	196	337	307	404	254	305	453
	PrecMin	534	43	22	1	2	5	7	16	11	27	7	20	24
	PrecStd	260	73	54	73	51	39	41	72	84	82	73	79	89
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	27.1	27.2	27.1	28.6	30.2	31.0	32.0	32.1	30.1	27.9	27.0
		MCWB	20.6	20.9	21.1	21.8	23.8	25.0	25.2	25.8	26.0	25.4	23.1	21.6
	2%	DB	24.1	24.5	24.8	25.7	27.5	28.9	30.0	31.1	31.0	29.1	27.2	25.8
		MCWB	20.7	20.0	20.4	21.4	22.7	24.3	24.6	25.5	25.5	24.5	22.8	21.5
	5%	DB	23.4	23.2	23.7	24.6	26.5	28.1	29.2	30.3	30.0	28.4	26.3	24.5
		MCWB	20.3	19.6	20.0	20.8	21.9	23.8	24.3	25.2	25.0	24.1	22.3	21.1
	10%	DB	22.8	22.6	22.7	23.8	25.8	27.7	28.8	29.7	29.1	27.7	25.6	23.8
		MCWB	19.8	19.3	19.6	20.3	21.7	23.6	24.2	24.9	24.7	23.7	21.9	20.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	22.6	22.7	23.5	24.8	25.8	26.1	26.8	26.8	27.1	25.0	23.6
		MCDB	24.0	24.3	24.4	24.5	26.8	28.4	28.6	29.9	29.8	28.0	27.0	25.2
	2%	WB	22.1	21.5	21.8	22.6	24.0	25.3	25.5	26.2	26.2	26.0	24.1	22.9
		MCDB	23.4	23.4	23.4	24.2	26.2	27.5	28.4	29.5	29.5	27.6	25.9	24.2
	5%	WB	21.4	20.8	21.2	22.1	23.3	24.8	25.1	25.7	25.7	25.1	23.5	22.2
		MCDB	22.8	22.6	22.8	23.7	25.4	27.1	28.2	29.1	28.9	27.3	25.4	23.7
	10%	WB	20.7	20.1	20.5	21.4	22.7	24.4	24.7	25.4	25.3	24.5	22.8	21.6
		MCDB	22.3	22.0	22.2	23.2	24.8	26.7	27.8	28.7	28.4	27.0	24.9	23.3
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.3	4.2	4.2	4.2	4.1	4.3	4.4	4.3	4.2	3.9	4.0
		MCDBR	4.9	5.1	4.8	4.6	4.7	4.5	4.7	4.9	5.1	4.6	4.4	4.5
		MCWBR	4.5	4.2	3.7	3.1	2.9	2.4	2.5	2.6	2.7	2.7	3.2	3.5
	5% WB	MCDBR	4.6	5.0	4.5	4.5	4.4	4.2	4.5	4.5	4.5	4.1	4.1	4.2
		MCWBR	4.9	4.5	4.0	3.5	3.0	2.3	2.6	2.6	2.6	2.7	3.3	3.8
(40) Clear-Sky Solar Irradiance	taub	0.356	0.364	0.385	0.412	0.410	0.393	0.379	0.378	0.376	0.369	0.346	0.356	(40)
	taud	2.571	2.542	2.470	2.402	2.456	2.546	2.608	2.620	2.613	2.616	2.665	2.604	(41)
	Ebn at Noon	888	909	906	887	884	893	904	906	903	893	896	872	(42)
	Edh at Noon	87	96	109	120	114	104	97	95	93	88	78	81	(43)
(44) All-Sky Solar Radiation	RadAvg	3.26	4.09	4.87	5.99	6.50	6.96	7.11	6.51	5.80	4.75	3.57	3.12	(44)
(45)	RadStd	0.26	0.28	0.32	0.33	0.41	0.42	0.37	0.40	0.29	0.31	0.23	0.24	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page